Automatically Generated Report

Conducted via Brain Researcher MCP and ClaudeCode

Multiverse Robustness of Resting-State Network Segregation in Cocaine Use Disorder

Same-Data Meta-Analysis with Generalized Least Squares across 36 Analytical Specifications

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Executive Summary

This report revises the cloned project summary using three local evidence sources: the result tables and scripts in the repository, the earlier Brain Researcher case-study PDF, and the full neuroimaging pipeline PDF. The core scientific story is a two-stage one. First, a single Schaefer-400 plus Tian Scale III pipeline reported strong positive associations between addiction severity and system segregation in Default, Striatum, Control, and HippAmyg networks. Second, the multiverse robustness layer asks whether the pre-specified findings survive plausible changes in atlas, motion threshold, and confound strategy.

The confirmatory robustness answer is negative: 0 rows in the confirmatory SDMA summary survive SDMA-GLS FDR correction. This means the original single-pipeline effects should be framed as analytically fragile findings, not as stable cocaine-use-disorder biomarkers.

The exploratory layer remains scientifically useful but must be labeled as hypothesis-generating: 7 all-network SDMA rows survive FDR correction in the local table, led by Thalamus, Visual, Control, Limbic, HippAmyg, and Dorsal Attention candidates. The revised claim boundary is therefore: robust evidence for specification sensitivity; candidate evidence for new aCompCor-sensitive or sub-threshold consensus effects.

Study Origin and Sample

The pipeline report identifies the source dataset as SUDMEX-CONN, OpenNeuro ds003346: a Mexican resting-state fMRI cohort including individuals with cocaine use disorder and healthy controls. The raw dataset is described as 144 BIDS subjects acquired on a 3T Siemens scanner at the National Institute of Psychiatry in Mexico City.

Quantity	**Value**	**Interpretation**
Raw BIDS dataset	144 subjects	SUDMEX-CONN / OpenNeuro ds003346.
Manual exclusions	4 subjects	Pre-processing exclusions for fieldmap failure or missing functional data.
Failed QC	6 subjects	Motion or coverage failures in the pipeline report.
Final analytic sample	134 subjects	Pipeline report final sample before analysis-specific missingness.
Groups on cover page	CUD 71, HC 63	Pipeline report cover summary.
Demographics table	CUD 71, HC 62	The same PDF table reports one fewer HC; this discrepancy is preserved here rather than silently resolved.
Multiverse result tables	max n = 138	The cloned result tables report n_subjects = 138 across analysis rows.

The Addiction Severity Index (ASI) is the key clinical predictor family for the dimensional CUD-only analyses. The pipeline report lists psychiatric, medical, family/social, drug, legal, and employment domains; the multiverse result table contains seven analysis labels: CUD-vs-HC plus six ASI-related labels. Because the local artifacts disagree on exact sample counts, this report preserves those differences and treats the present work as a rendered evidence synthesis rather than a new subject-level audit.

Pipeline and Atlas

The original pipeline is not just a statistical table. It starts from BIDS data and proceeds through fMRIPrep, Rapidtide, and XCP-D before extracting parcellated time series and connectivity features. This context matters because the original single-specification findings are tied to a specific denoising and parcellation workflow.

Stage	Version / setting	Role
fMRIPrep	24.0.0	Minimal preprocessing, MNI152NLin2009cAsym 2 mm output, fsLR CIFTI output, FreeSurfer reconstruction, confound TSVs.
Rapidtide	2.9.6	Systemic low-frequency oscillation estimation and removal using gray-matter restricted lagged regressors.
XCP-D	0.7.4	Nuisance regression, CIFTI processing, FD censoring, smoothing, and parcellated time series.
Base denoising	36P plus sLFO	36-parameter model with global signal regression plus Rapidtide custom confound.
Sensitivity denoising	24P and aCompCor	Additional confound strategies introduced for the multiverse robustness layer.

Component	Value	Scientific role
Cortex	Schaefer-400	Yeo-7 cortical network labels.
Subcortex	Tian Scale III	50 validated subcortical regions, including striatum, thalamus, hippocampus, and amygdala subdivisions.
Combined single-spec atlas	450 ROIs	Pipeline report concatenates Schaefer-400 and Tian Scale III time series.
Multiverse atlases	6 atlas choices	Schaefer-200/Tian III; Schaefer-400/Tian III; Schaefer-600/Tian III; Schaefer-400/HCP; Gordon/Tian III; Glasser/Tian III.

The pipeline report emphasizes Tian Scale III because it separates subcortical systems that are relevant to addiction, including nucleus accumbens shell/core, putaminal zones, thalamic nuclei, and hippocampal/amygdala subdivisions. The multiverse layer later broadens this atlas decision into six atlas specifications.

Feature and Model Definitions

For each subject, the pipeline report describes a combined FC matrix and three feature families. System segregation is the central readout in the multiverse report because it turns within- and between-network connectivity into one interpretable network self-containment score.

Feature family	Count	Definition
Functional connectivity	55 features	10 within-network and 45 between-network Fisher-z-averaged correlations.
System segregation	10 features	One SyS value per network: within-network connectivity minus between-network connectivity, normalized by absolute between-network connectivity.
Graph theory	14 features	Top-20-percent thresholded FC graph summarized with modularity, global efficiency, clustering, and participation metrics.

The single-pipeline statistical model is an OLS regression for each feature: outcome feature as a function of predictor, age, sex, and mean framewise displacement. A1 uses binary CUD-vs-HC group; A3 uses continuous ASI composite scores within CUD participants. Benjamini-Hochberg FDR correction is applied within each feature set and analysis.

Original Single-Pipeline Findings

The pipeline report's core result is not a null result. Under the original Schaefer-400 plus Tian Scale III, 36P plus Rapidtide/XCP-D workflow, the A3 dimensional ASI analyses find five FDR-significant positive SyS associations. Higher addiction severity is interpreted as greater network self-containment.

Analysis	Network	Beta	t	p	pFDR	Circuit
ASI medical	Default	+74.53	5.13	.000003	.000032	Self-referential / DMN
ASI medical	Striatum	+18.57	3.39	.001	.006	Reward / habit
ASI medical	Control	+13.36	2.92	.005	.016	Cognitive control
ASI family	Default	+83.56	3.87	.0003	.003	Self-referential / DMN
ASI drug	HippAmyg	+1.33	3.06	.003	.033	Memory / cue reactivity

The strongest original signal is Default Mode Network segregation with ASI medical severity ($t = 5.13$, pFDR = .000032). The pipeline interpretation links DMN hyper-segregation to drug-focused rumination and somatic/interoceptive burden; Striatum to reward/habit circuitry; Control to frontoparietal regulation; and HippAmyg to memory, emotional context, and cue reactivity.

Multiverse Robustness Design

The multiverse combines atlas, framewise-displacement threshold, and confound-regression choices. Each specification estimates systemic segregation for network-outcome pairs and then aggregates specification-level Z statistics with Same-Data Meta-Analysis using Generalized Least Squares.

Dimension	Options in bundle
Atlas	Schaefer-200/Tian III; Schaefer-400/Tian III; Schaefer-600/Tian III; Schaefer-400/HCP; Gordon/Tian III; Glasser/Tian III
Motion threshold	Documented SDMA subset uses FD 0.3 mm and FD 0.5 mm; master table also contains 36P FD 0.4 rows.
Confound strategy	36P with global signal regression; 24P without GSR; aCompCor without GSR
Outcome family	7 analysis labels x 10 systemic-segregation features
SDMA matrix C	36 x 36; off-diagonal mean 0.153, min -0.322, max 0.937

The local master table has 42 specification IDs and 2940 rows because it includes six additional 36P FD=0.4 specifications. The documented 36-specification SDMA subset has 36 specification IDs and 2520 rows. This distinction is important: the repo contains more rows than the named 36-specification report, but the SDMA figures and summary table are aligned to the 36-specification subset.

Confirmatory Robustness Verdict

Across the confirmatory summary table, 0 rows have SDMA-GLS FDR $q < 0.05$. This is the central robustness result: the originally reported single-pipeline associations fail to survive multiverse aggregation.

Outcome	GLS Z	p	FDR q	Stouffer Z	Stouffer p	Sig specs
DMN SyS ~ ASI Medical (CUD)	0.99	0.323	0.388	1.36	0.173	4/36
DMN SyS ~ ASI Family (CUD)	1.01	0.311	0.388	0.79	0.428	3/36
HippAmyg SyS ~ ASI Drug (CUD)	1.19	0.234	0.388	2.01	0.044	7/36
DMN SyS ~ CUD vs HC	-0.03	0.980	0.980	0.31	0.756	0/36
SalVentAttn SyS ~ ASI Employment (CUD)	1.79	0.074	0.388	2.11	0.035	4/36
Cont SyS ~ ASI Employment (CUD)	1.51	0.131	0.388	2.26	0.024	2/36

Outcome	36P sig	24P sig	aCompCor sig
DMN SyS ~ ASI Medical (CUD)	2/12	2/12	0/12
DMN SyS ~ ASI Family (CUD)	3/12	0/12	0/12
HippAmyg SyS ~ ASI Drug (CUD)	4/12	2/12	1/12
DMN SyS ~ CUD vs HC	0/12	0/12	0/12
SalVentAttn SyS ~ ASI Employment (CUD)	0/12	2/12	2/12
Cont SyS ~ ASI Employment (CUD)	0/12	2/12	0/12

This section changes the scientific interpretation of the original result. The single-pipeline analysis remains useful as the motivating discovery, but the multiverse layer says the five pre-specified findings are not stable enough to be treated as validated biomarkers.

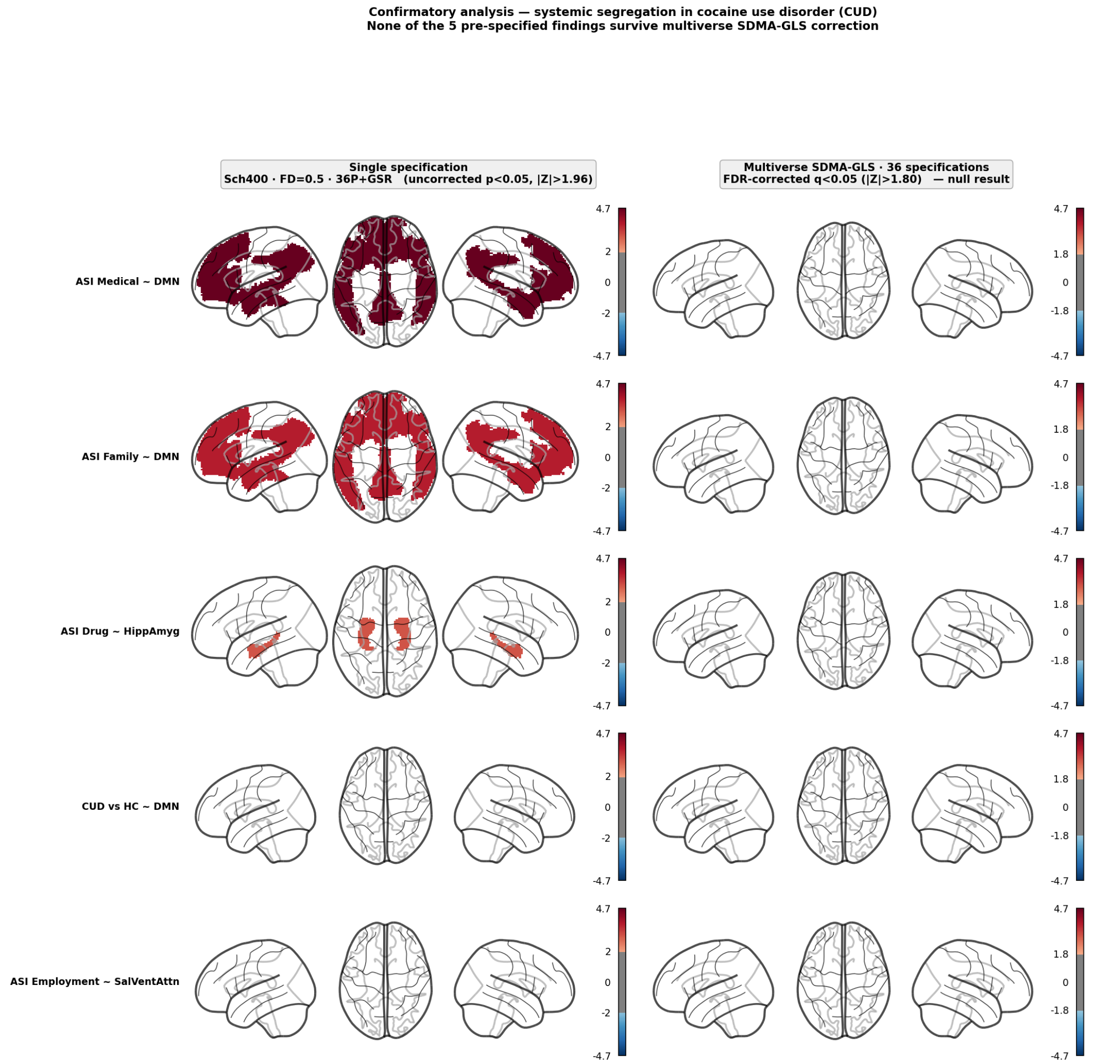

Figure 1: Confirmatory figure from the cloned artifact bundle. The top row reflects the original single specification; the SDMA-GLS row shows that the confirmatory effects do not survive the multiverse aggregation.

Exploratory Candidate Signals

The all-network SDMA table contains 15 rows with absolute GLS Z >= 1.96. After FDR correction across the all-network family, 7 rows remain below q < 0.05. These do not rescue the original confirmatory claims, but they define a clearer next-study target set.

Analysis	Feature	GLS Z	q	36P sig	24P sig	aCompCor sig
A3_ASI_medical_sys	sys_Thalamus	3.88	0.007	0/12	0/12	5/12
A3_ASI_employment_sys	sys_Vis	3.40	0.024	0/12	0/12	3/12
A3_ASI_drug_sys	sys_Thalamus	3.29	0.024	0/12	0/12	6/12
A3_ASI_legal_sys	sys_Cont	3.20	0.024	0/12	2/12	1/12
A3_ASI_family_sys	sys_Limbic	3.12	0.026	0/12	2/12	0/12
A3_ASI_medical_sys	sys_HippAmyg	2.95	0.037	0/12	1/12	1/12
A1_CUD_vs_HC_sys	sys_DorsAttn	2.89	0.039	0/12	0/12	0/12

The earlier case-study PDF highlights two key exploratory patterns: thalamic SyS with ASI medical and ASI drug, and Dorsal Attention SyS for CUD-vs-HC. The first pattern is especially tied to aCompCor specifications; the second illustrates an SDMA feature where consistent sub-threshold effects can yield a consensus signal even when individual specifications do not pass FDR.

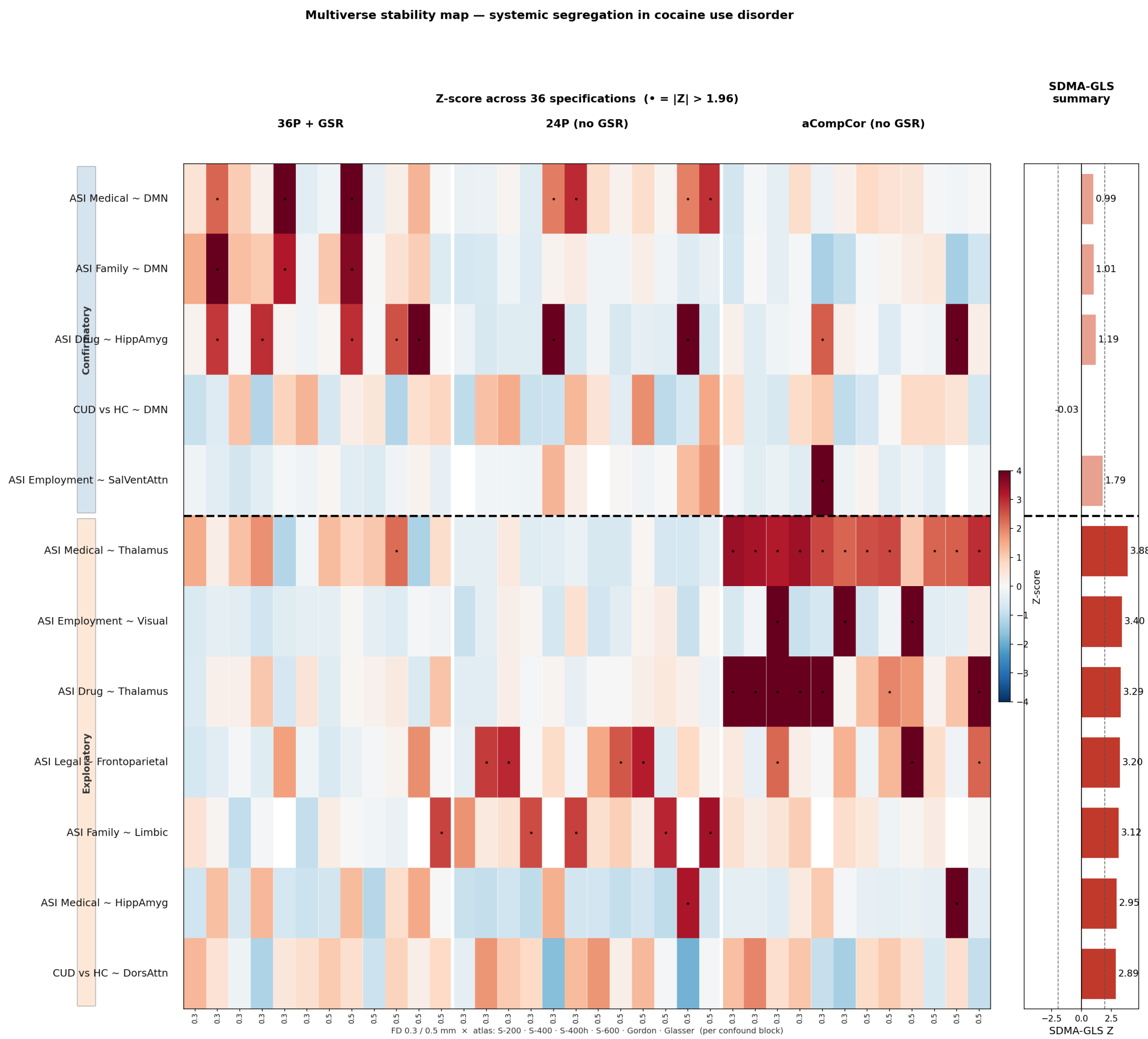

Figure 2: Stability map showing specification-level Z scores across network-outcome pairs. Columns are sorted by confound strategy, making the original GSR sensitivity and the exploratory aCompCor-sensitive signals visible.

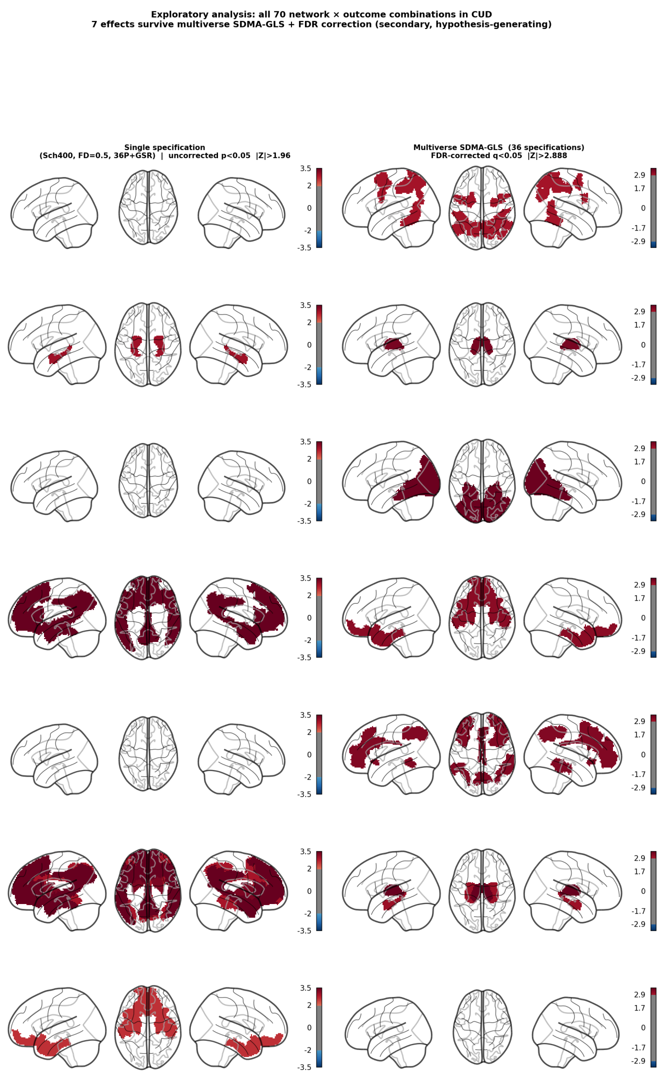

Figure 3: All-network SDMA comparison figure. Exploratory signals should be treated as candidates because the primary confirmatory analysis remains null.

Interpretation

The strongest interpretation is methodological. Resting-state CUD brain-behaviour associations can look convincing in a single atlas/denoising/motion configuration, but the direction and strength of the result can shift materially when plausible preprocessing and analysis choices are varied.

The biological interpretation should be layered. The original pipeline suggests a coherent Default-Striatum-Control-HippAmyg hyper-segregation story for addiction severity. The robustness audit does not prove that story false; it shows that the specific pre-specified effects are not stable enough across the multiverse. The exploratory thalamic and dorsal-attention candidates should therefore be treated as replication targets rather than final discoveries.

The role of global signal regression is central but not one-dimensional. Some original DMN effects cluster in 36P/GSR specifications, whereas ASI Employment-SalVentAttn is stronger without GSR and the thalamic exploratory signals are strongest in aCompCor. This is why the report should avoid a simple claim that GSR alone causes or removes every effect.

Limitations and Artifact Boundaries

This BR-template report was regenerated from files present in the cloned repository. The raw SUD-MEX imaging data and XCP-D derivatives are not included, so this step verifies report rendering and table consistency, not a full neuroimaging rerun.

There are source-level inconsistencies that matter for scientific framing: the pipeline PDF reports 144 raw subjects, 134 after QC, and group counts that differ slightly between the cover and demographics table; the multiverse tables report 138 subjects in the subject-count field. The pipeline PDF reports a 450-ROI Schaefer-400 plus Tian Scale III construction, while some text in the feature extraction section refers to a 456 x 456 matrix. The repository report is also a derived-result bundle, not a complete BIDS derivatives release.

These inconsistencies do not invalidate the robustness conclusion, but they should prevent overclaiming. A publication-grade handoff should reconcile sample counts, analysis-specific missingness, exact atlas dimensionality, and whether each multiverse specification was recomputed from raw derivatives or from post-hoc reparameterized outputs.

Reproducibility Bundle

Path	Shape/status	Role
master_summary_all.tsv	2940 rows	All specification-level tests and covariates.
sdma_summary.tsv	6 rows	Confirmatory SDMA-GLS/Stouffer summary table.
inter_spec_correlation.tsv	36 x 36	Inter-specification correlation matrix C.
brain_figures/confirmatory_figure.png	figure	Primary confirmatory brain summary.
brain_figures/stability_map.png	figure	Z-score stability map across specifications.
brain_figures/sdma_comparison_all.png	figure	All-network SDMA comparison map.
Brain_Researcher_case2_Cocaine_Use.pdf	23 pages	Earlier BR scientific report containing the core multiverse interpretation.
14_pipeline_report (2).pdf	12 pages	Pipeline report from BIDS through fMRIPrep, Rapidtide, XCP-D, FC, SyS, and original single-specification results.
scripts/preprocessing/*.sh	9 scripts	BIDS-to-XCP-D operational pipeline skeleton.
scripts/multiverse/*.py	7 scripts	Multiverse, SDMA, brain-figure, and report generation code.

Generated outputs are `results/br\_scientific\_report.tex`, `results/br\_scientific\_report.pdf`, and `results/combined\_report.pdf`. The combined PDF appends the original Matplotlib multiverse report and the two source PDFs after the revised BR-template scientific report.

References

Lefort-Besnard, J., Nichols, T. E., and Maumet, C. (2025). Statistical inference for same data meta-analysis in neuroimaging multiverse analyses. Imaging Neuroscience.

Angeles-Valdez, D. et al. (2022). The Mexican magnetic resonance imaging dataset of patients with cocaine use disorder: SUDMEX CONN. Scientific Data, 9, 133.

Mehta, K. et al. (2024). XCP-D: A robust pipeline for the post-processing of fMRI data. Imaging Neuroscience, 2, 1-26.